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Sulphide Capacities in Some Multi Component Slag Systems

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Experimental measurements have been carried out to determine the sulphide capacities of some slags in the Al_2O_3–MnO, Al_2O_3–MgO–MnO, Al_2O_3–CaO–MgO–MnO, Al_2O_3–CaO–MgO–SiO_2, Al_2O_3–CaO–MnO–SiO_2, and Al_2O_3–CaO–MgO–MnO–SiO_2 systems in the temperature range of 1 773–1 923 K. The gas–slag equilibration technique has been employed using the gas mixtures of Ar–CO–CO_2–SO_2. Repetition of the measurements has revealed very good reproducibility of the experimental results. The measured sulphide capacities have been used to examine the reliability of a sulphide capacity model developed earlier in the present laboratory. The model predictions based on the model parameters up to ternary interactions obtained in the corresponding lower order systems have shown good agreement with the experimental data. Model parameters necessary for the calculation of sulphide capacities of the above mentioned systems have been provided.

KEY WORDS: sulphide capacity; slags; sulphide capacity model; Al_2O_3–CaO–MgO–MnO–SiO_2.

1. Introduction

In view of the extreme importance of sulphide capacities of metallurgical slags, an effort is currently being made at the Division of Theoretical Metallurgy to get a reliable description of sulphide capacities in the multicomponent system, Al_2O_3–CaO–Cr_2O_3–FeO–Fe_2O_3–MgO–MnO–SiO_2, which covers most slags used in the pyrometallurgical industries. A sulphide capacity model was developed,[1,2] so that sulphide capacity, C_S in a given system could be described as a function of both temperature as well as slag composition. A series of experimental studies have been carried out in this laboratory to determine the sulphide capacities of binary and ternary slags.[3–8] Based on the experimental data obtained by the researchers in the present laboratory and other research groups, model calculations were made for various binary and ternary oxide systems. The application of the model to these systems has shown that the model is successful in interpolating and extrapolating experimental data. Since the main goal of the model development is to be able to predict the sulphide capacities of slags containing more than three components using solely the information from the corresponding lower order systems, it is necessary to make model calculations for higher-order systems and check the results against experimental data. A trial test was made earlier.[9] In that exercise, the sulphide capacities in some high-order systems were predicted based on the available binary and ternary interaction parameters of the model. However, some model parameters were still missing at that time due to lack of experimental information in the corresponding binary and ternary systems. While the comparison of the calculated sulphide capacities with the experimental results were promising, it was still not evident to make a conclusive remark regarding the reliability of the model in predicting sulphide capacities for high-order systems. It was felt that experimental data for some binary and ternary systems had to be generated before the model could be put into real test. A number of experimental studies have since then been carried out for selected ternary systems.[7,8] Based on the results of these experiments and previous studies, a set of model parameters have been obtained.

The aim of the present work is threefold, *viz.* (1) experimental determination of sulphide capacities of some slags in the Al_2O_3–MnO, Al_2O_3–MgO–MnO, Al_2O_3–CaO–MgO–MnO, Al_2O_3–CaO–MgO–SiO_2, Al_2O_3–CaO–MnO–SiO_2, and Al_2O_3–CaO–MgO–MnO–SiO_2 systems, (2) model predictions of the sulphide capacities of these slags from the information of lower order systems, (3) examination of the reliability of the sulphide capacity model in predicting C_S of multicomponent slgas.

2. Experimental

The gas–slag equilibration method was employed in the present study. The experimental setup and procedure have been described in detail in earlier publications.[4,10] The oxides and gases used in the present work are listed in **Table 1**. The oxide powders except MnO were calcined at 1 273 K for 10 h before being mixed thoroughly in the required proportions. Five grams of each powder mixture was prefused at 1 873 K in a platinum crucible. The prefused slags were ground to fine powders and stored in a desicator before use. The Ar, CO, CO_2 and SO_2 gases were purified before mixing. A detailed description of the gas cleaning and gas-controlling systems have also been presented in the earlier paper.[4,10]

In a general run, after the prefused slag was remelted and homogenized at the experimental temperature, the $CO-CO_2-SO_2-Ar$ gas mixture was introduced into the reaction chamber. A constant flow rate of 400 ml/min was maintained for at least 6 h to assure the equilibrium between the slag and the gas. The equilibrium partial pressures of different gases in the gas phase calculated by the Gibbs energy minimization program SOLGASMIX[11] are presented in **Table 2**. The Gibbs energies of the reactions involved were taken from the SGTE (Scientific Group Thermodata Europe) databank. The samples kept in a sample holder were pulled out to the cold end of the furnace and cooled down under the flow of the gas mixture. The sample was visually examined to ascertain that it was completely molten at the

Table 1. The oxides and gases used in the present work.

Material	Purity
Aluminium Oxide, (Al_2O_3)	
Calcium Oxide, (CaO)	Anhydrous, Analytical Reagent Grade
Magnesium Oxide, (MgO)	pro analyse grade
Manganese Oxide, (MnO)	Spec.pure (99.5% purity)
Silicon Oxide, (SiO_2)	pro analyse grade
Argon, (Ar.)	Argon Plus (Ar>99.99%)
Carbon monoxide, (CO)	S grade
Carbon dioxide, (CO_2)	S grade
Sulphur dioxide, (SO_2)	S grade

Table 2. Equilibrium partial pressures of various gaseous species at the experimental temperatures.

Species	Partial Pressure (atm.)			
	1923 K	1873 K	1823 K	1773 K
Ar	$5.029\cdot10^{-1}$	$5.039\cdot10^{-1}$	$5.017\cdot10^{-1}$	$5.024\cdot10^{-1}$
CO	$2.245\cdot10^{-1}$	$2.186\cdot10^{-1}$	$1.604\cdot10^{-1}$	$1.553\cdot10^{-1}$
CO_2	$2.277\cdot10^{-1}$	$2.344\cdot10^{-1}$	$2.907\cdot10^{-1}$	$2.965\cdot10^{-1}$
SO_2	$3.508\cdot10^{-2}$	$3.233\cdot10^{-2}$	$4.158\cdot10^{-2}$	$3.913\cdot10^{-2}$
S_2	$5.305\cdot10^{-3}$	$6.941\cdot10^{-3}$	$2.814\cdot10^{-3}$	$4.201\cdot10^{-3}$
O_2	$4.613\cdot10^{-7}$	$2.034\cdot10^{-7}$	$2.176\cdot10^{-7}$	$8.557\cdot10^{-8}$

Table 3. Experimental data and sulphide capacities of Al_2O_3–MnO slags.

Sample	Temp. [K]	Composition		$\sqrt{\frac{p_{S_2}}{p_{O_2}}}$	mass % S	C_S (expt.)	C_S (average)
		X_{Al2O3}	X_{MnO}				
A1	1923	0.257	0.723	107	0.225 0.209	$2.104\cdot10^{-3}$ $1.960\cdot10^{-3}$	$2.032\cdot10^{-3}$
A2	1923	0.243	0.757	107	0.318 0.303	$2.971\cdot10^{-3}$ $2.830\cdot10^{-3}$	$2.900\cdot10^{-3}$
A3	1923	0.233	0.767	107	0.526 0.495	$4.917\cdot10^{-3}$ $4.622\cdot10^{-3}$	$4.769\cdot10^{-3}$
A4	1923	0.217	0.783	107	0.616 0.569	$5.756\cdot10^{-3}$ $5.319\cdot10^{-3}$	$5.558\cdot10^{-3}$
A1	1873	0.257	0.723	185	0.152 0.183 0.146	$8.230\cdot10^{-4}$ $9.910\cdot10^{-4}$ $7.910\cdot10^{-4}$	$8.683\cdot10^{-4}$
A2	1873	0.243	0.757	185	0.276 0.246	$1.490\cdot10^{-3}$ $1.320\cdot10^{-3}$	$1.405\cdot10^{-3}$
A3	1873	0.233	0.767	185	0.459 0.405 0.498	$2.486\cdot10^{-3}$ $2.190\cdot10^{-3}$ $2.690\cdot10^{-3}$	$2.455\cdot10^{-3}$
A4	1873	0.217	0.783	185	0.923 0.819	$4.990\cdot10^{-3}$ $4.430\cdot10^{-3}$	$4.710\cdot10^{-3}$
A1	1823	0.257	0.723	114	0.081	$7.090\cdot10^{-4}$	$7.090\cdot10^{-4}$
A2	1823	0.243	0.757	114	0.126 0.125	$1.108\cdot10^{-3}$ $1.100\cdot10^{-3}$	$1.104\cdot10^{-3}$
A3	1823	0.233	0.767	114	0.222 0.177	$1.950\cdot10^{-3}$ $1.550\cdot10^{-3}$	$1.750\cdot10^{-3}$
A4	1823	0.217	0.783	114	0.373 0.404	$3.270\cdot10^{-3}$ $3.550\cdot10^{-3}$	$3.410\cdot10^{-3}$
A1	1773	0.257	0.723	222	0.122 0.112	$5.484\cdot10^{-4}$ $5.050\cdot10^{-4}$	$5.267\cdot10^{-4}$
A3	1773	0.233	0.767	222	0.395 0.364	$1.780\cdot10^{-3}$ $1.640\cdot10^{-3}$	$1.710\cdot10^{-3}$
A4	1773	0.217	0.783	222	0.559 0.593	$2.517\cdot10^{-3}$ $2.670\cdot10^{-3}$	$2.594\cdot10^{-3}$

Table 4. Experimental data and sulphide capacities of Al_2O_3–MgO–MnO slags.

Sample	Temp. [K]	Composition			$\sqrt{\frac{p_{S_2}}{p_{O_2}}}$	mass % S	C_S (expt.)	C_S (average)
		X_{Al2O3}	X_{MgO}	X_{MnO}				
B1	1923	0.235	0.113	0.652	107	1.301 1.278	$1.216\cdot10^{-2}$ $1.194\cdot10^{-2}$	$1.205\cdot10^{-2}$
B2	1923	0.207	0.148	0.645	107	1.369 1.402	$1.279\cdot10^{-2}$ $1.310\cdot10^{-2}$	$1.294\cdot10^{-2}$
B3	1923	0.193	0.159	0.648	107	1.541 1.659	$1.440\cdot10^{-02}$ $1.650\cdot10^{-2}$	$1.545\cdot10^{-2}$
B4	1923	0.161	0.253	0.586	107	1.657 1.685	$1.929\cdot10^{-2}$ $1.836\cdot10^{-2}$	$1.883\cdot10^{-2}$
B1	1873	0.235	0.113	0.652	185	1.511 1.469	$8.167\cdot10^{-3}$ $7.940\cdot10^{-3}$	$8.054\cdot10^{-3}$
B4	1873	0.61	0.253	0.586	185	2.866 2.914	$1.549\cdot10^{-2}$ $1.575\cdot10^{-2}$	$1.562\cdot10^{-2}$

Table 5. Experimental data and sulphide capacities of Al_2O_3–CaO–MgO–MnO slags.

Sample	Temp. [K]	Composition			$\sqrt{\frac{p_{S_2}}{p_{O_2}}}$	mass % S	C_S (expt.)	C_S (average)
		X_{Al2O3}	X_{CaO}	X_{MgO}				
C1	1923	0.289	0.556	0.086	107	0.382 0.446	$3.570\cdot10^{-3}$ $4.164\cdot10^{-3}$	$3.867\cdot10^{-3}$
C2	1923	0.305	0.557	0.087	107	0.293 0.217	$2.739\cdot10^{-3}$ $2.025\cdot10^{-3}$	$2.382\cdot10^{-3}$
C3	1923	0.299	0.538	0.121	107	0.316 0.322	$2.952\cdot10^{-3}$ $3.013\cdot10^{-3}$	$2.983\cdot10^{-3}$
C4	1923	0.301	0.550	0.121	107	0.244 0.226	$2.280\cdot10^{-3}$ $2.111\cdot10^{-3}$	$2.196\cdot10^{-3}$
C1	1873	0.289	0.556	0.086	185	0.546 0.509	$2.950\cdot10^{-3}$ $2.756\cdot10^{-3}$	$2.853\cdot10^{-3}$
C2	1873	0.305	0.557	0.087	185	0.462 0.451	$2.499\cdot10^{-3}$ $2.438\cdot10^{-3}$	$2.469\cdot10^{-3}$
C3	1873	0.299	0.538	0.121	185	0.427 0.399	$2.306\cdot10^{-3}$ $2.157\cdot10^{-3}$	$2.232\cdot10^{-3}$
C4	1873	0.301	0.550	0.087	185	0.360 0.347	$1.948\cdot10^{-3}$ $1.875\cdot10^{-3}$	$1.912\cdot10^{-3}$

experimental temperature. The sulphur content of the slag was determined by the stoichiometric combustion method[12] using a METROHM automatic titration unit.

3. Results

Sulphide capacities in the Al_2O_3–MnO, Al_2O_3–MgO–MnO, Al_2O_3–CaO–MgO–MnO, Al_2O_3–CaO–MgO–SiO_2, Al_2O_3–CaO–MnO–SiO_2, and Al_2O_3–CaO–MgO–MnO–SiO_2 systems were determined in the temperature range of 1 773–1 923 K. A number of slag compositions were chosen for each system. The experimental results for these systems are presented in **Tables 3–8** respectively. It should be mentioned that the compositions listed in these tables are the weighed-in compositions. Previous study[4] showed that the results of chemical analysis of the slag samples were in good agreement with the weighed-in compositions except for a slight decrease in manganese content. Most of the experiments were repeated. It is seen in Tables 3–8 that good reproducibility was obtained for almost all the measurements. The average values of C_S are also listed in the same tables. Sharma and Richardson[13] have measured the sulphide capacities of some Al_2O_3–MnO slags at 1 923 K using the gas mixtures of CO_2–H_2–SO_2–N_2. Their results are compared with the present experimental data in **Fig. 1**. It seems that the C_S values obtained by these authors are considerably higher than the results of the present work.

To the knowledge of the present authors, among the other systems studied with present work, only the sul-

Table 6. Experimental data and sulphide capacities of Al_2O_3–CaO–MgO–SiO_2 slags.

Sample	Temp. [K]	Composition X_{Al2O3}	X_{CaO}	X_{MgO}	$\sqrt{\frac{p_{S_2}}{p_{O_2}}}$	mass % S	C_S (expt.)	C_S (average)
Q1	1923	0.118	0.178	0.224	107	0.0226 0.0229	$2.109\cdot10^{-4}$ $2.140\cdot10^{-4}$	$2.125\cdot10^{-4}$
Q2	1873	0.096	0.433	0.151	185	0.0758 0.0730	$4.099\cdot10^{-4}$ $3.948\cdot10^{-4}$	$4.024\cdot10^{-4}$
Q3	1873	0.049	0.378	0.145	185	0.0249 0.0214	$1.344\cdot10^{-4}$ $1.157\cdot10^{-4}$	$1.251\cdot10^{-4}$
Q4	1873	0.068	0.359	0.214	185	0.0512 0.0441	$2.768\cdot10^{-4}$ $2.383\cdot10^{-4}$	$2.576\cdot10^{-4}$
Q2	1823	0.096	0.433	0.151	114	0.0225 0.0239	$1.977\cdot10^{-4}$ $2.098\cdot10^{-4}$	$2.038\cdot10^{-4}$
Q3	1823	0.049	0.378	0.145	114	0.0146	$1.280\cdot10^{-4}$	$1.280\cdot10^{-4}$
Q5	1773	0.085	0.404	0.216	222	0.0390 0.0417	$1.757\cdot10^{-4}$ $1.882\cdot10^{-4}$	$1.819\cdot10^{-4}$
Q2	1773	0.096	0.433	0.151	222	0.0301	$1.357\cdot10^{-4}$	$1.357\cdot10^{-4}$
Q3	1773	0.049	0.378	0.145	222	0.0114 0.0100	$5.149\cdot10^{-5}$ $4.512\cdot10^{-5}$	$4.831\cdot10^{-5}$
Q4	1773	0.068	0.359	0.214	222	0.0182	$8.185\cdot10^{-5}$	$8.185\cdot10^{-5}$

Table 7. Experimental data and sulphide capacities of Al_2O_3–CaO–MnO–SiO_2 slags.

Sample	Temp. [K]	Composition X_{Al2O3}	X_{CaO}	X_{MnO}	$\sqrt{\frac{p_{S_2}}{p_{O_2}}}$	mass % S	C_S (expt.)	C_S (average)
L1	1923	0.131	0.119	0.093	107	0.0088 0.0082	$8.249\cdot10^{-5}$ $7.672\cdot10^{-5}$	$7.961\cdot10^{-5}$
L2	1923	0.205	0.125	0.148	107	0.0124 0.0114 0.0101	$1.155\cdot10^{-4}$ $1.069\cdot10^{-4}$ $9.419\cdot10^{-5}$	$1.055\cdot10^{-4}$
L1	1873	0.131	0.119	0.093	185	0.0114 0.0136	$6.143\cdot10^{-5}$ $7.338\cdot10^{-5}$	$6.741\cdot10^{-5}$
L2	1873	0.205	0.125	0.148	185	0.0143 0.0144	$7.732\cdot10^{-5}$ $7.773\cdot10^{-5}$	$7.753\cdot10^{-5}$
L1	1823	0.131	0.119	0.093	114	0.0035 0.0034	$3.033\cdot10^{-5}$ $2.962\cdot10^{-5}$	$2.996\cdot10^{-5}$
L1	1773	0.131	0.119	0.093	222	0.0019	$8.344\cdot10^{-6}$	$8.344\cdot10^{-6}$
L2	1773	0.205	0.125	0.148	222	0.0052 0.0048	$2.363\cdot10^{-5}$ $2.162\cdot10^{-5}$	$2.263\cdot10^{-5}$

phide capacities in the quaternary Al_2O_3–CaO–MgO–SiO_2 system have been experimentally determined earlier.[14–17] While Tsao and Katayama[17] determined sulphide capacities of high Al_2O_3 containing slags at three temperatures using slag–metal equilibration technique, the other works were carried out at 1 773 K using the gas–slag equilibration technique. The sulphide capacity values reported by different authors[14–16] are compared with the present results at 1 773 K in **Table 9**. It is seen that while the values obtained by Kärsrud[16] and Kalyanram *et al.*[15] follow the same trend as the present results, the sulphide capacities suggested by Abraham and Richardson[14] are considerably higher than the data of the other researchers. It should be mentioned that Kalyanram *et al.*[15] have studied 27 slag compositions. Table 9 only includes the C_S for four slags, the compositions of which are in the same region as the present work. However, all the C_S values reported by these authors[15] appear to follow the same trend. A further comparison will be given in the model calculation part of this paper.

Table 8. Experimental data and sulphide capacities of Al_2O_3–CaO–MgO–MnO–SiO_2 slags.

Sample	Temp. [K]	Composition				$\sqrt{\frac{p_{S_2}}{p_{O_2}}}$	mass % S	C_S (expt.)	C_S (average)
		X_{Al2O3}	X_{CaO}	X_{MgO}	X_{MnO}				
S1	1923	0.084	0.073	0.351	0.041	107	0.016 0.015	$1.456\cdot10^{-4}$ $1.375\cdot10^{-4}$	$1.416\cdot10^{-4}$
S2	1923	0.114	0.072	0.359	0.082	107	0.023 0.019	$2.157\cdot10^{-4}$ $1.733\cdot10^{-4}$	$1.945\cdot10^{-4}$
S3	1873	0.064	0.449	0.075	0.026	185	0.054 0.035	$2.897\cdot10^{-4}$ $1.873\cdot10^{-4}$	$2.385\cdot10^{-4}$
S4	1873	0.063	0.388	0.132	0.036	185	0.059 0.064	$3.129\cdot10^{-4}$ $3.448\cdot10^{-4}$	$3.288\cdot10^{-4}$
S1	1873	0.084	0.073	0.351	0.041	185	0.011 0.012	$5.934\cdot10^{-5}$ $6.573\cdot10^{-5}$	$6.254\cdot10^{-5}$
S2	1873	0.114	0.072	0.359	0.082	185	0.013 0.013	$6.998\cdot10^{-5}$ $6.619\cdot10^{-5}$	$6.808\cdot10^{-5}$
S4	1773	0.063	0.388	0.132	0.036	222	0.037 0.037	$1.672\cdot10^{-4}$ $1.526\cdot10^{-4}$	$1.599\cdot10^{-4}$
S1	1773	0.084	0.073	0.351	0.041	222	0.007 0.008	$3.293\cdot10^{-5}$ $3.453\cdot10^{-5}$	$3.373\cdot10^{-5}$
S2	1773	0.114	0.072	0.359	0.082	222	0.009	$3.895\cdot10^{-5}$	$3.895\cdot10^{-5}$

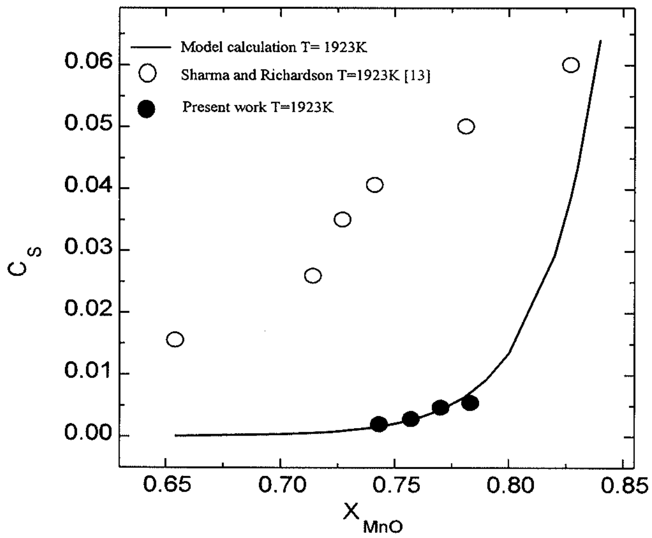

Fig. 1. Sulphide capacity as function of composition in the Al_2O_3–MnO system.

4. Model Calculation

As mentioned in the introduction part, one of the tasks of the present work is to predict the sulphide capacities in the Al_2O_3–MnO, Al_2O_3–MgO–MnO, Al_2O_3–CaO–MgO–MnO, Al_2O_3–CaO–MgO–SiO_2, Al_2O_3–CaO–MnO–SiO_2, and Al_2O_2–CaO–MgO–MnO–SiO_2 systems. The predicted C_S need to be compared with the experimental data, so that the reliability of the model in predicting sulphide capacities of multicomponent slags can be examined. The use of sulphide capacity model in the case of oxide systems has been discussed in detail in an earlier work.[2] The model is only briefly outlined in this section to provide an orientation for the readers.

According to the definition of sulphide capacity, C_S can be expressed as:

$$C_S=\exp\left(\frac{-\Delta G^\circ}{RT}\right)\left(\frac{a_{O^{2-}}}{f_{S^{2-}}}\right)=\exp\left(\frac{-\Delta G^\circ}{RT}\right)\exp\left(\frac{\xi}{RT}\right) \quad \text{.........................(1)}$$

where ΔG° is the Gibbs energy change of the following reaction:

$$\frac{1}{2}S_2(g)+O^{2-}(\text{slag})=S^{2-}(\text{slag})+\frac{1}{2}O_2(g) \quad \text{....(2)}$$

In the sulphide capacity model,[2] ξ is taken as zero in the case of pure liquid FeO. Based on the experimental data for the sulphide capacity of FeO,[16,18] ΔG° of Reaction (2) is evaluated to be:

$$\Delta G^\circ=118\,535-58.8157\cdot T \quad (\text{J/mol}) \quad \text{..........(3)}$$

In the model, the ratio of the activity of O^{2-} to the activity coefficient of S^{2-}, $a_{O^{2-}}/f_{S^{2-}}$ is expressed as

$$\frac{a_{O^{2-}}}{f_{S^{2-}}}=\exp\left(-\frac{\xi}{RT}\right) \quad \text{......................(4)}$$

where R is the gas constant, T is temperature in K. In the case of unary systems, ξ in a function of temperature. On the other hand, ξ in a multicomponent system is described as a function of both temperature and composition,

$$\xi=\sum X_i\xi_i+\xi_{\text{mix}} \quad \text{.........................(5)}$$

In Eq. (5), the subscript i denotes component i, X_i is the mole fraction of this component. While the first term in this equation represents the "linear" variation of ξ from

Table 9. A comparison of the present experimental results with the literature data in the system Al_2O_3–CaO–MgO–SiO_2.

Authors	Composition			C_S
	X_{Al2O3}	X_{CaO}	X_{MgO}	
Abraham & Richardson [14]	0.086	0.448	0.174	$5.43\cdot10^{-4}$
	0.116	0.457	0.177	$6.85\cdot10^{-4}$
	0.151	0.492	0.152	$6.92\cdot10^{-4}$
	0.188	0.637	0.079	$3.97\cdot10^{-3}$
Kärsrud [16]	0.071	0.458	0.074	$1.29\cdot10^{-4}$
	0.068	0.337	0.213	$1.05\cdot10^{-4}$
	0.069	0.396	0.145	$1.15\cdot10^{-4}$
Kalyanram *et. al*[15]	0.077	0.332	0.150	$3.86\cdot10^{-5}$
	0.074	0.390	0.146	$1.68\cdot10^{-4}$
	0.071	0.309	0.214	$5.86\cdot10^{-5}$
	0.105	0.318	0.148	$2.70\cdot10^{-4}$
Present work	0.085	0.404	0.216	$1.82\cdot10^{-4}$
	0.096	0.433	0.151	$1.36\cdot10^{-4}$
	0.049	0.378	0.145	$4.83\cdot10^{-5}$
	0.068	0.359	0.214	$8.19\cdot10^{-5}$

the pure components in the absence of the interaction between different species, ξ_{mix} takes account of the mutual interactions between different species. This term is described in terms of polynomials in both temperature and composition.[2] As mentioned earlier, the sulphide capacity model is expected to predict sulphide capacities of multicomponent slags using the model parameters up to ternary interactions. In order to make reliable model predictions, the relevant binary and ternary interaction parameters need to be optimized using well determined sulphide capacity data in the corresponding binary and ternary systems. In order to meet such requirements, a series of experimental studies have been carried out earlier in the present laboratory to measure the sulphide capacities of various binary and ternary slags.[4–9] Based on the results of these previous experimental studies and experimental data available in literature,[14,19–21] a set of model parameters that are necessary for the estimation of the sulphide capacities in the quaternary and quinary systems studied in the present work have been optimized. These parameters as well as the references corresponding to the earlier work are presented in **Table 10**. It is noted that the number of digits of the model parameters does not reflect the accuracy of the experimental data. In this table, y_i stands for the cation fraction of cation i, defined as the number of the cations of i over the total number of the cations:

$$y_i=\frac{N_i}{\sum_{j=1}^{n} N_j} \quad\text{..............................(6)}$$

On the basis of the slag composition, the cation fractions can be calculated and the value of ξ of a multicomponent slag can thus be evaluated using parameters listed in Table 10. To help the readers use the model, the expression of ξ for the Al_2O_3–CaO–MgO–MnO–SiO_2 slag is presented below as an example:

$$\begin{aligned}\xi= & X_{Al_2O_3}\xi_{Al_2O_3}+X_{CaO}\xi_{CaO}+X_{MgO}\xi_{MgO}+X_{MnO}\xi_{MnO}\\ & +X_{SiO_2}\xi_{SiO_2}+\xi^{Al_2O_3-CaO}_{interaction}+\xi^{Al_2O_3-SiO_2}_{interaction}+\xi^{Al_2O_3-MnO}_{interaction}\\ & +\xi^{CaO-SiO_2}_{interaction}+\xi^{MgO-SiO_2}_{interaction}+\xi^{MnO-SiO_2}_{interaction}+\xi^{Al_2O_3-CaO-MgO}_{interaction}\\ & +\xi^{Al_2O_3-CaO-SiO_2}_{interaction}+\xi^{Al_2O_3-MgO-SiO_2}_{interaction}+\xi^{Al_2O_3-MgO-MnO}_{interaction}\\ & +\xi^{Al_2O_3-MnO-SiO_2}_{interaction}+\xi^{CaO-MgO-SiO_2}_{interaction}+\xi^{CaO-MnO-SiO_2}_{interaction}\\ & +\xi^{MgO-MnO-SiO_2}_{interaction} \quad\text{..............................(7)}\end{aligned}$$

Equation (7) can also be used for the calculation of ξ in lower order systems, in which case the mole fractions of the species that are absent and their cation fractions are set to zero.

The calculated results for Al_2O_3–MnO at 1 923 K along with present experimental data and those reported earlier[13] are presented in Fig. 1. Model calculations were carried out for the quaternary and quinary slags that were also experimentally studied. A comparison between the results of model prediction and the present experimental data is presented in **Fig. 2**. The experimentally determined C_S values in the CaO–MgO–MnO–SiO_2 system that were determined earlier in the present laboratory[7] are also included in this figure. Even the experimental data obtained for the Al_2O_3–CaO–MgO–SO_2 system by other researchers[15,16] are included. It is seen that in general, the model predictions for all the multicomponent systems are in good agreement with the experimental data. The experimental data reported by Kalyanram *et al.*[15] and Kärsrud[16] are also reasonably well predicted by the model calculations. However, it should be mentioned that a few experimental values are considerably higher than the model predictions. This disagreement needs be sorted out by further

Table 10. Optimized model parameters.

Unary	ξ_i	Ref.
Al_2O_3	$1.57705276 \cdot 10^5$	Ref. (2)
CaO	$-3.3099425 \cdot 10^4$	Ref. (2)
MgO	$9.57307326 \cdot 10^3$	Ref. (3)
MnO	$-3.66264629 \cdot 10^4$	Ref. (2)
SiO_2	$1.68872587 \cdot 10^5$	Ref. (2)
Binary interaction	$\xi_{\text{interaction}}$	
Al_2O_3–CaO	$\xi_{\text{interaction}}^{Al_2O_3-CaO} = y_{Al^{3+}} \cdot y_{Ca^{2+}} \cdot [9.82827968 \cdot 10^4 + 55.07340941 \cdot T]$	Ref. (5)
Al_2O_3–MnO	$\xi_{\text{interaction}}^{Al_2O_3-MnO} = y_{Al^{3+}} \cdot y_{Mn^{2+}} \cdot [4.33987403 \cdot 10^5 + 2.49121594 \cdot 10^2 \cdot T \cdot (y_{Al^{3+}} - y_{Mn^{2+}})]$	Present work
Al_2O_3–SiO_2	$\xi_{\text{interaction}}^{Al_2O_3-SiO_2} = y_{Al^{3+}} \cdot y_{Si^{4+}} \cdot [1.86850468 \cdot 10^5]$	Ref. (6)
CaO–SiO_2	$\xi_{\text{interaction}}^{CaO-SiO_2} = y_{Ca^{2+}} \cdot y_{Si^{4+}} \cdot [9.72717695 \cdot 10^4 + 72.8749746 \cdot T]$	Ref. (7)
MgO–SiO_2	$\xi_{\text{interaction}}^{MgO-SiO_2} = y_{Mg^{2+}} \cdot y_{Si^{4+}} \cdot [6.97403222 \cdot 10^5 - 2.24084556 \cdot 10^2 \cdot T]$	Ref. (3)
MnO–SiO_2	$\xi_{\text{interaction}}^{MnO-SiO_2} = y_{Mn^{2+}} \cdot y_{Si^{4+}} \cdot [-3.22911470 \cdot 10^5 + 2.12029980 \cdot 10^2 \cdot T + 1.34860658 \cdot 10^5 \cdot (y_{Mn^{2+}} - y_{Si^{4+}})]$	Ref. (7)
Ternary interaction	$\xi_{\text{interaction}}$	Ref.
Al_2O_3–CaO–MgO	$\xi_{\text{interaction}}^{Al_2O_3-CaO-MgO} = y_{Al^{3+}} \cdot y_{Ca^{2+}} \cdot y_{Mg^{2+}} \cdot [4.1659555 \cdot 10^6 - 1.0665663 \cdot 10^3 \cdot T - 3.04080189 \cdot 10^6 \cdot y_{Al^{3+}}]$	Ref. (8)
Al_2O_3–CaO–SiO_2	$\xi_{\text{interaction}}^{Al_2O_3-CaO-SiO_2} = y_{Al^{3+}} \cdot y_{Ca^{2+}} \cdot y_{Si^{4+}} \cdot [-2.03579264 \cdot 10^6 + 6.86044695 \cdot 10^2 \cdot T]$	Ref. (6)
Al_2O_3–MgO–MnO	$\xi_{\text{interaction}}^{Al_2O_3-MgO-MnO} = y_{Al^{3+}} \cdot y_{Mg^{2+}} \cdot y_{Mn^{2+}} \cdot [-1.56149723 \cdot 10^6 + 2.72278645 \cdot 10^3 \cdot T - 1.22741846 \cdot 10^7 \cdot y_{Al^{3+}}]$	Present work
Al_2O_3–MgO–SiO_2	$\xi_{\text{interaction}}^{Al_2O_3-MgO-SiO_2} = y_{Al^{3+}} \cdot y_{Ag^{2+}} \cdot y_{Si^{4+}} \cdot [1.56192588 \cdot 10^5 - 2.90498555 \cdot 10^2 \cdot T + 9.49447247 \cdot 10^5 \cdot y_{Al^{3+}}]$	Ref. (8)
Al_2O_3–MnO–SiO_2	$\xi_{\text{interaction}}^{Al_2O_3-MnO-SiO_2} = y_{Al^{3+}} \cdot y_{Mn^{2+}} \cdot y_{Si^{4+}} \cdot [1.56584846 \cdot 10^6 - 6.62494162 \cdot 10^2 \cdot T - 5.32290311 \cdot 10^6 \cdot y_{Al^{3+}}]$	Ref. (8)
CaO–MgO–SiO_2	$\xi_{\text{interaction}}^{CaO-MgO-SiO_2} = y_{Ca^{2+}} \cdot y_{Mg^{2+}} \cdot y_{Si^{4+}} \cdot [-1.52649771 \cdot 10^6 + 6.25663842 \cdot 10^2 \cdot T + 1.48525598 \cdot 10^6 \cdot y_{Ca^{2+}}]$	Ref. (3)
CaO–MnO–SiO_2	$\xi_{\text{interaction}}^{CaO-MnO-SiO_2} = y_{Ca^{2+}} \cdot y_{Mn^{2+}} \cdot y_{Si^{4+}} \cdot [-1.17989159 \cdot 10^6 + 6.21243714 \cdot 10^2 \cdot T - 1.19111179 \cdot 10^6 \cdot y_{Ca^{2+}}]$	Ref. (7)
MgO–MnO–SiO_2	$\xi_{\text{interaction}}^{MgO-MnO-SiO_2} = y_{Mg^{2+}} \cdot y_{Mn^{2+}} \cdot y_{Si^{4+}} \cdot [9.10360927 \cdot 10^6 - 4.42600708 \cdot 10^3 \cdot T - 2.86966462 \cdot 10^6 \cdot y_{Mg^{2+}}]$	Ref. (9)
Al_2O_3–CaO–MnO	No experimental data	
CaO–MgO–MnO	No experimental data	

experimentation.

The calculated iso-sulphide capacity lines for the Al_2O_3–CaO–MgO–MnO, Al_2O_3–CaO–MgO–SiO_2, Al_2O_3–CaO–MnO–SiO_2 and Al_2O_3–CaO–MgO–MnO–SiO_2 systems at 1 873 K are presented in **Figs. 3–6** respectively. It should be pointed out that no phase diagram information is available for these systems except for the Al_2O_3–CaO–MgO–SiO_2 quaternary. Sections of phase diagrams at constant mass percentages of MgO are found in the literature.[22] Hence, the section for this system shown in Fig. 4 is given on the basis of weight percent. In the case of other systems, the iso-sulphide capacity lines are plotted in the sections constructed on the basis of mole fractions. The phase boundaries given in Figs. 3, 5, 6 are only tentative. The experimental results obtained for these sections are also included in these figures for comparison. The agreement between the predicted values and the experimental data can be considered satisfactory in the case of all the systems.

5. Discussion

In Fig. 1, the calculated sulphide capacity as function of composition at 1 923 K in the Al_2O_3–MnO system is compared with the experimental data obtained in the present work as well as those reported by Sharma and Richardson.[13] The results show that the C_S data from the present study are well reproduced by the calculated sulphide capacity curves. The results also reveal that a high MnO content leads to a slag with high sulphide capacity.

As shown in Figs. 2 to 6, model predictions, in general, agree well with the experimentally determined sulphide

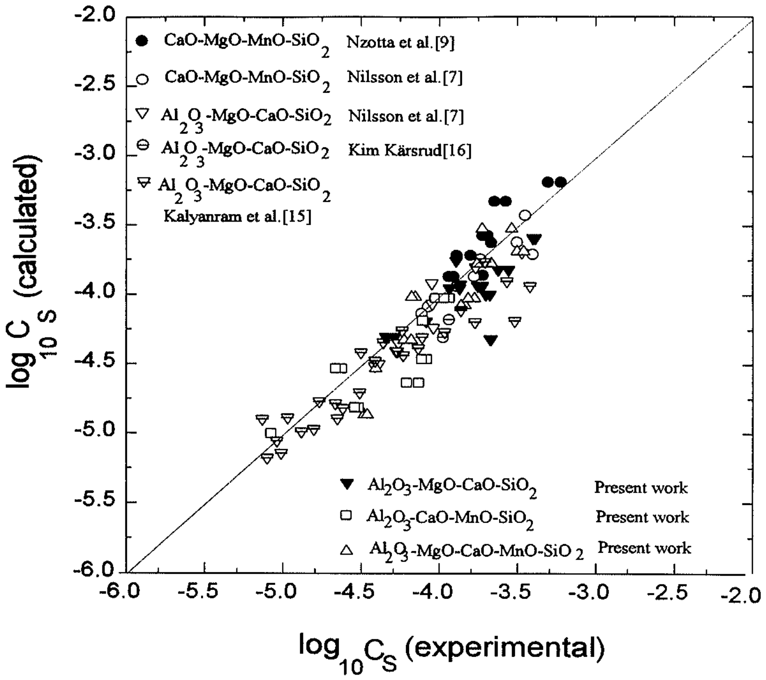

Fig. 2. Comparison between the results of the model predictions and the experimental data.

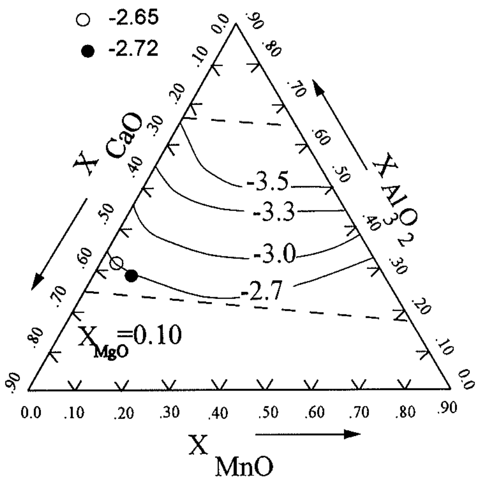

Fig. 3. Iso-sulphide capacity lines (in terms of logarithm) for Al_2O_3–CaO–MgO–MnO system at T=1 873 K with X_{MgO}=0.10.

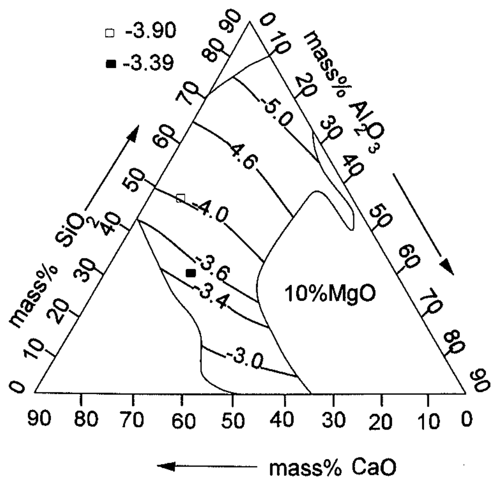

Fig. 4. Iso-sulphide capacity lines (in terms of logarithm) for Al_2O_3–CaO–MgO–SiO_2 system at T=1 873 K with mass% MgO=0.1.

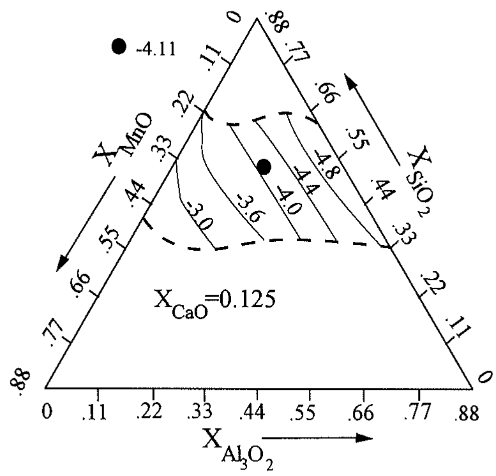

Fig. 5. Iso-sulphide capacity lines (in terms of logarithm) for Al_2O_3–CaO–MnO–SiO_2 system at T=1 873 K with X_{CaO}=0.125.

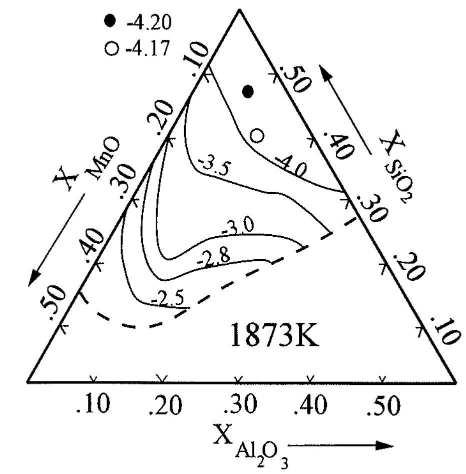

Fig. 6. Iso-sulphide capacity lines (in terms of logarithm) for Al_2O_3–CaO–MgO–MnO–SiO_2 system at T=1 873 K with X_{CaO}=0.072, X_{MgO}=0.355.

capacity data in the case of all multicomponent oxide systems studied. It is reasonable to conclude that Eq. (7) along with the model parameters listed in Table 10 provides a useful tool for the estimation of sulphide capacity of any slag in the Al_2O_3–CaO–MgO–MnO–SiO_2 system.

The iso-sulphide capacity contours in the case of Al_2O_3–CaO–MgO–MnO at $X_{MgO}=0.10$ and $T=1\,873$ K are presented in Fig. 3. That slags with higher MnO content have high C_S values is well brought out in this figure.

Slags containing Al_2O_3, CaO, MgO and SiO_2 are very important in the iron and steel making industry. The slags used for ladle refining usually contain less than 10% SiO_2 and about 50 mass% CaO. Figure 4 shows that such slags have very high sulphide capacities. The iso-sulphide capacity lines in Fig. 4 indicate that at a constant MgO content (Mass%MgO = 10), C_S increases dramatically with the increase of the CaO content in the Al_2O_3–CaO–MgO–SiO_2 system. On the other hand, the replacement of SiO_2 by Al_2O_3 has no strong effect on the ability of the slag to capture sulphur.

In Fig. 5, the iso-C_S contours in the Al_2O_3–CaO–MnO–SiO_2 system at $X_{CaO}=0.125$ and 1 873 K show that MnO has very high affinity for sulphur. This aspect is also demonstrated in Fig. 6, where the iso-sulphide capacity lines are plotted at 1 837 K for the Al_2O_3–CaO–MgO–MnO–SiO_2 system with $X_{CaO}=0.072$ and $X_{MgO}=0.355$. A high MnO content leads to a slag with high C_S.

The effect of the replacement of Al_2O_3 by CaO on the sulphide capacity at constant MgO and SiO_2 contents (7.5 mass% MgO, 7.5 mass% SiO_2) is shown in **Fig. 7**. It is seen that the sulphide capacity increases with the CaO content irrespective of the temperature. An increase of X_{CaO} from 0.50 to 0.56 leads to an increase of 100% in C_S at steel making temperature. This effect is more profound in the slags with high CaO contents. It can be expected that the variation of the compositions of the slag might have strong impact on the ability of the slag to take up sulphur.

The good agreement between the model predictions and the experimental data also indicates that the sulphide capacity model[2] developed in the present laboratory can be successfully used in predicting sulphide capacities for multicomponent slags, when the model parameters up to ternary interactions are available. It is felt that most of the future experimental work should be devoted to the measurements of sulphide capacities in binary and ternary systems, so that the corresponding binary and ternary interaction parameters can be optimized on a sound basis. The sulphide capacities of the slags covered by the Al_2O_3–CaO–Cr_2O_3–FeO–Fe_2O_3–MgO–MnO–SiO_2 system are expected to be well predicted.

It should be mentioned that there are still some experimental data, although only few, from which the model predictions show slight deviation. The reason for the disagreement could be due to either the experimental uncertainties or the model parameters obtained in the studies of lower order systems. It is seen in Table 10 that some binary and ternary interactions are absent. The lack of these parameters might be another reason for the disagreement of the few data points. The absent parameters can be classified into two groups. The first group contains the interactions between cations of basic oxides like Ca^{2+}, Mn^{2+} and Mg^{2+}. In view of the high liquidus temperatures in the corresponding binary or ternary systems, it was not possible to carry out C_S measurements of these slags. Hence no attempt has been made to optimize these parameters. However, it is reasonable to assume that the interactions between these cations are negligible due to their similarity. The second group includes the interactions involving Al^{3+} and one or two cations of basic oxides, for example the Al^{3+}–Ca^{2+}–Mn^{2+} ternary interactions. In spite of this, the model does provide a useful tool for the estimation of C_S of multicomponent slags. The present model would also be an aid to process simulation and controlling, where the sulphide capacity need to be described as a function of temperature and composition. In the case, where some of the binary or ternary parameters are missing or in doubt, it would always be wise to make one or two spot-checks by experimental measurements.

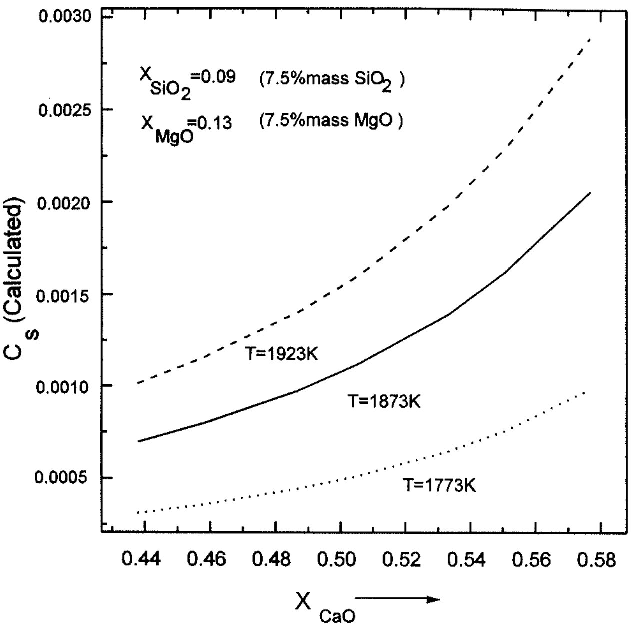

Fig. 7. The effect of the replacement of Al_2O_3 by CaO on sulphide capacity at constant MgO (7.5 mass%) and SiO_2 (7.5 mass%) contents.

6. Summary

The sulphide capacities of some multicomponent slags in the Al_2O_3–MnO, Al_2O_3–MgO–MnO, Al_2O_3–CaO–MgO–MnO, Al_2O_3–CaO–MgO–SiO_2, Al_2O_3–CaO–MnO–SiO_2, and Al_2O_3–CaO–MgO–MnO–SiO_2 systems have been determined using gas–slag equilibration method. Model predictions of the sulphide capacities for the same slags have been made using the model parameter up to ternary interactions obtained in the corresponding lower order systems. Comparison of the model predictions with the experimental results has shown that the sulphide capacity model developed in the present

laboratory can be successfully used in predicting sulphide capacities for multicomponent slags, when the model parameters up to ternary interactions are available. An expression has been suggested to estimate the sulphide capacities of the slags in the Al_2O_3–CaO–MgO–MnO–SiO_2 system.

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